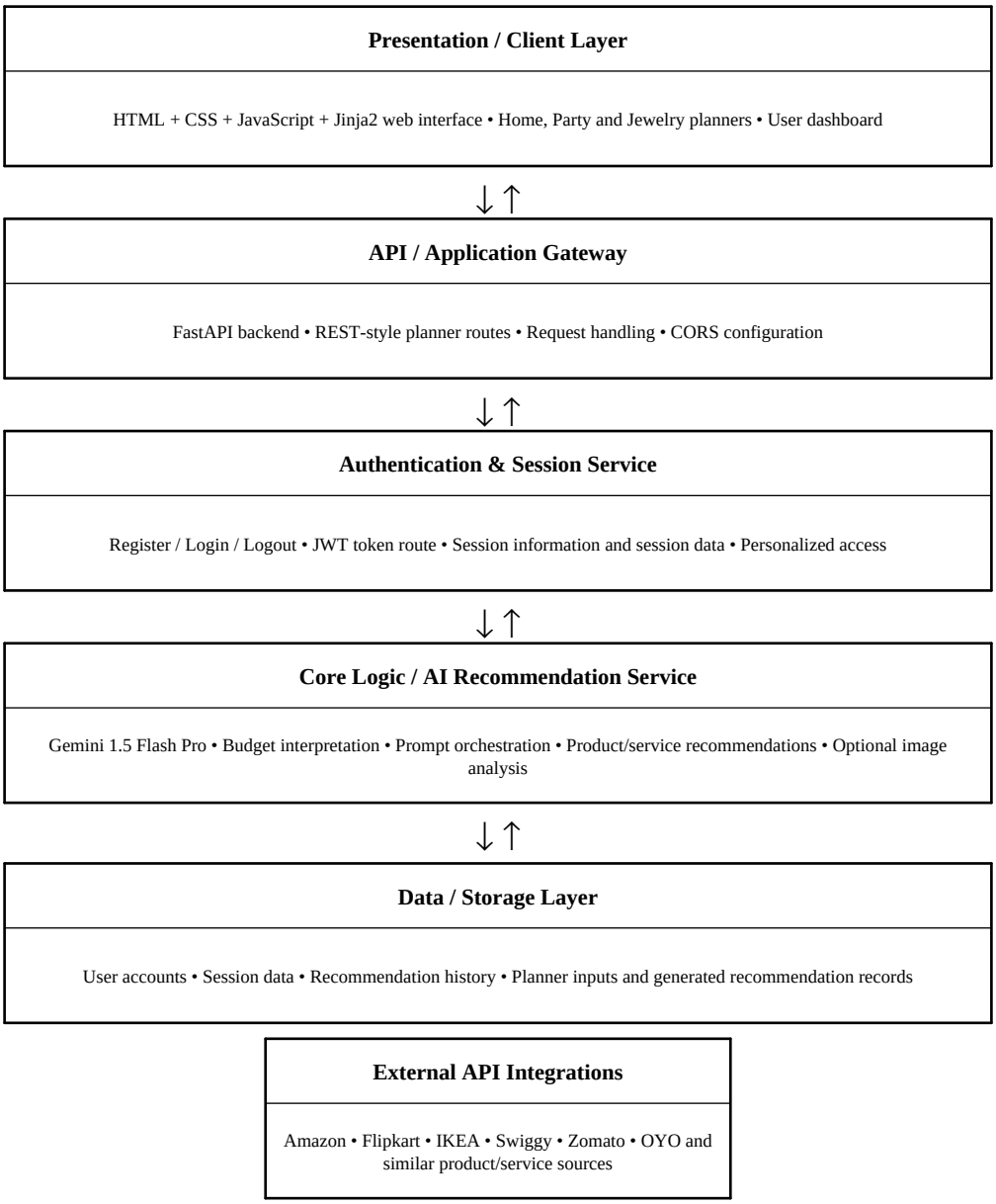
Solution Architecture

Date	01 October 2026
Team ID	SWTID-2026-5883
Project Name	Pocket Smart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram

The architecture below follows the supplied Solution Architecture template and maps Pocket Smart AI into presentation, application, integration, and data/storage components.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects user budgets, preferences, planner details and optional outfit images; displays AI-generated recommendations in a structured interface.	HTML, CSS, JavaScript, Jinja2
API / Application Layer	Receives planner requests, routes them to the correct module, communicates with the AI service, and returns recommendation results.	FastAPI, Python, Uvicorn
Authentication & Session	Handles registration, login, logout, JWT token generation, current-session information and personalized session data.	FastAPI routes, JWT, session handling
Core AI / Recommendation Service	Interprets budget and context, builds prompts, generates recommendations, handles category-specific logic, and supports text plus optional image inputs.	Gemini 1.5 Flash Pro, Python
Planner Modules	Home Interior Planner recommends furniture/decor/lighting; Party Planner handles event budget, guests, food, venue and decoration; Jewelry Planner considers occasion, style and optional outfit image.	FastAPI routes, Gemini prompts, HTML/Jinja2
External Integrations	Provides product and service information used to create platform-aware suggestions. Development documentation also describes mock API calls or simulated scraping.	Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO and similar sources
Data / Storage	Supports user-specific session data, recommendation history, planner inputs and generated results for review or reuse.	Application data/session storage; exact database technology is not specified in the supplied project material

Team Work Allocation

S.No.	Name	Assigned Architecture Work
1	Jashvini.K	AI & Recommendation Architecture — Gemini integration, prompt orchestration, budget interpretation, recommendation generation and optional image-analysis flow.
2	Indhumathi.D	Backend Architecture — FastAPI routes, authentication, JWT/session handling, CORS and API communication.
3	Jayapriya Rajkumar	Presentation Architecture — HTML/CSS/JavaScript/Jinja2 UI, planner forms, result cards and responsive interface.
4	Jonal.A	Integration & Planner Modules — Home, Party and Jewelry workflows plus external product/service platform integration.
5	Kavibharathi.K	Data, Testing & Documentation — recommendation history/session data, validation, testing, optimization and final architecture documentation.

This document follows the structure of the supplied Solution Architecture template: a high-level architecture diagram followed by a component description table.
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